```
% Caleb Distel | SES 598
% 11/17/2025
% Chirp Radar matlab activity
clear; clc; close all;

Fs   = 60e6;
T    = 5e-4; % chirp duration 20e-6
t    = 0:1/Fs:T-1/Fs;
f1   = 10e6; f2 = 20e6;
fLO = 15e6; % mix for diff: -5...5; sum: 25...35
M    = 2; % Decimate by 2 Fs -> Fs2 = 60MHz -> 30MHz
Nfft = 8192;
f_xlim = [-40 40];
% Next: mix -> LPF -> decimate
```

Build signal and mix LO

```
x      = chirp(t, f1, T, f2, 'linear'); % sweep 10 -> 20 MHz
lo     = cos(2 *pi*fLO*t); % 15 MHz LO
y_mix = x .* lo;
```

LPF to satisfy new nyquist for Fs2 (30MHz) -> Low Pass filter to accept < 15MHz

```
N = 96; Fc = 8e6; % anti-alias LPF to keep -5...5 MHz
b = fir1(N, Fc/(Fs/2)); % Hamming LPF
y_lp = filter(b, 1, y_mix); % filtered signal
```

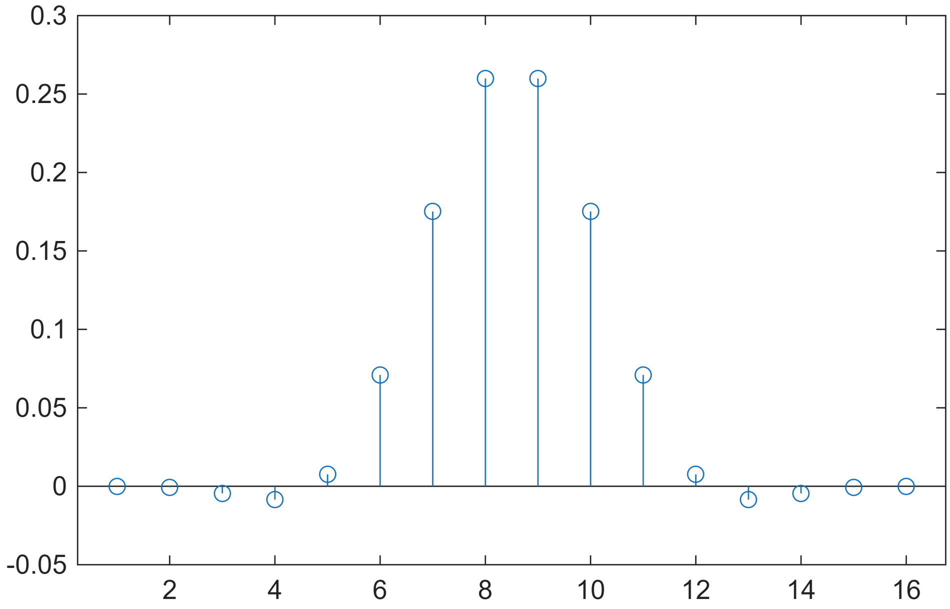
Decimate with M = 2 to drop every other sample (Fs -> Fs2)

```
y_ds = y_lp(1:M:end); % downsample by 2
Fs2  = Fs/M;          % Fs -> Fs2 (60 MHz -> 30MHz)
```

LPF + blackman filter

```
L = 15;

Wn = Fc/(Fs/2);
win = blackman(L+1);
h = fir1(L, Wn, "low", win, "scale");
hf = fi(h, 1, 12, 11);
stem(h);
```



Stage Spectra

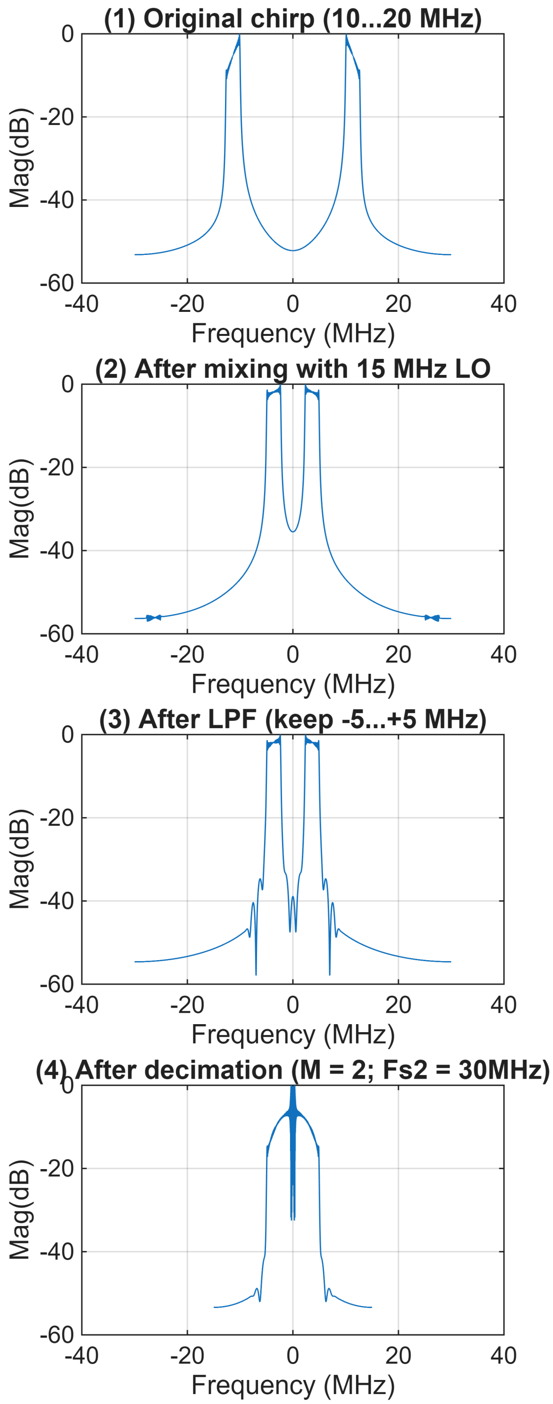
```
[fx, XdB] = spec(x,     Fs,  Nfft, h);  % original signal
[fm, MdB] = spec(y_mix, Fs,  Nfft, h);  % mixed signal (x + LO)
[ff, LdB] = spec(y_lp,  Fs,  Nfft, h);  % Filtered signal -> mixed[ f < 15MHz ]
[fd, DdB] = spec(y_ds,  Fs2, Nfft, h); % Decimated signal -> filtered @ 30MHz (Fs2)

figure('Color', 'w', 'Position', [70 70 1020 740*4]);
subplot(4,1,1); plot(fx, XdB); grid on; xlim(f_xlim);
title('(1) Original chirp (10...20 MHz)'); xlabel('Frequency (MHz)');
ylabel('Mag(dB)');

subplot(4,1,2); plot(fm, MdB); grid on; xlim(f_xlim);
title('(2) After mixing with 15 MHz LO'); xlabel('Frequency (MHz)');
ylabel('Mag(dB)');

subplot(4,1,3); plot(ff, LdB); grid on; xlim(f_xlim);
title('(3) After LPF (keep -5...+5 MHz)'); xlabel('Frequency (MHz)');
ylabel('Mag(dB)');

subplot(4,1,4); plot(fd, DdB); grid on; xlim(f_xlim);
title('(4) After decimation (M = 2; Fs2 = 30MHz)'); xlabel('Frequency (MHz)');
ylabel('Mag(dB)');
```



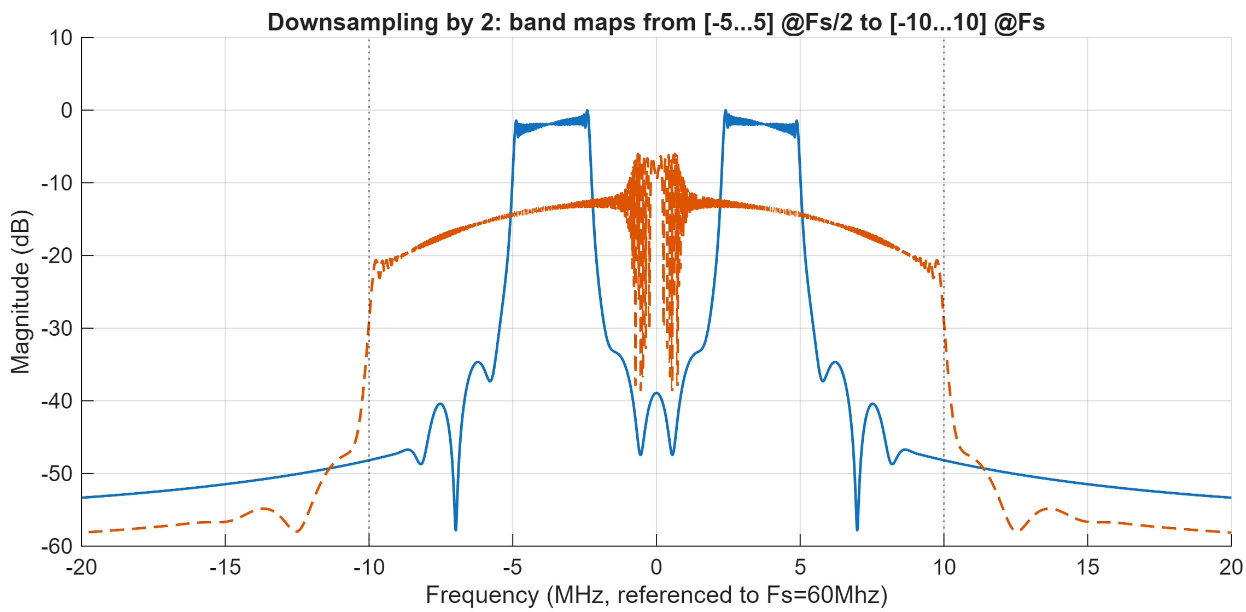
Compare on original-Fs (60MHz) axis (-10...+10 MHz)

```
[ff0, Ypre_dB]  = spec(y_lp, Fs,  Nfft, h);  % Pre-decimation (Fs)
```

```
[fd2, Ypost_dB] = spec(y_ds, Fs2, Nfft, h); % Post-decimation (Fs2)

fd2_on_Fs       = fd2 * M;       % Map to original Fs (*M)
Ypost_on_Fs_dB = Ypost_dB - 6; % amplitude is half -> ~-6 dB

figure('Color', 'w', 'Position', [120 120 900 380]);
hold on; grid on;
plot(ff0, Ypre_dB, 'Linewidth', 1.2);
plot(fd2_on_Fs, Ypost_on_Fs_dB, '--', 'Linewidth', 1.2);
xline([-10 10], ':k');
xlim([-20 20]); xlabel('Frequency (MHz, referenced to Fs=60Mhz)');
ylabel('Magnitude (dB)');
title('Downsampling by 2: band maps from [-5...5] @Fs/2 to [-10...10] @Fs');
```




```
% legend('Before decimation (after LPF)', 'After decimation (mapped to Fs, -6db at
M = 2)', 'Location', 'SouthWest');
```

Generate chirp table

```
clocks_per_T = T*Fs
```

```
clocks_per_T =
30000
```

```
chirp_counts = 0:clocks_per_T
```

```
chirp_counts = 1×30001
     0     1     2     3     4     5     6     7     8     9    10    11    12 ···
```

```
chirp_outputs = linspace(f1, f2, length(chirp_counts))
```

```
chirp_outputs = 1×30001
```

```
10^7 ×
    1.0000    1.0000    1.0001    1.0001    1.0001    1.0002    1.0002    1.0002 ⋯
```

Spectrum plotter function

```
function [f_MHz, XdB] = spec(x, Fs, Nfft, h)
    x = x(:);                       % column vector

    xw = conv(x, h, 'same');

    X = fft(xw, Nfft);

    X     = fftshift(X);
    f_Hz  = (-Nfft/2:Nfft/2-1) * (Fs/Nfft);
    f_MHz = f_Hz / 1e6;

    magX = abs(X);
    magX = magX ./ max(magX + eps);
    XdB  = 20*log10(magX + eps);
end
```